



CSE6324-SOFTWARE VERIFICATION & VALIDATION

Topic: Hospital Management System

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Test Summary Report

Hospital Management System

(HMS)

Independent Verification and Validation

Project Title: HMS Independent Verification and Validation	
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Document Control

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Name	Position	Signature
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Version History

Version	Release Date	Section	Amendments
1.0	22September,2019	All	Original Document

Distribution List

Version	Release Date	Controlled Copy No	Recipient Name	Department	Issue Date	Return Date
1.0.0	22/09/2019	01		HMS	22/09/2019	

Test Summary Report

General Information	
Test Report Description	The Test Summary Report is used to summarize the results of functional testing in Hospital Management System (HMS). The report covers test execution status, completion status of test cases, pass/fail status of tests, incident status, severity of defects, coverage, and final recommendations.
Version Author	CHIN ZHEN HO
Associated Test Plan Reference	HMS_TP
Contact Number	N/A

Summary

Test Status Report Summary

Test Status Item	Number
Number of Test Cases Planned to be Completed	74
Number of Test Cases Remaining to be Executed	0
Number of Test Cases Completed	74

The testing activities for HMS were completed based on the planned functional test cases from F001 to F019. The test execution covered User Authentication, Patient Functions, Doctor Functions, and Admin Functions. The test cases were executed using manual testing and tool-supported testing.

Current Status of Test Incident Report

New	Number of New Incidents = 3
Open	Number of Open Incidents = 3
Rejected	Number of Rejected Incidents = 0
Resolved	Number of Resolved Incidents = 0
Deferred	Number of Deferred Incidents = 0

During the test execution three incidents were recorded. All three are still open as the corresponding defect fixes or confirmations of retesting were not finalized during the current testing cycle.

Incident List

Incident ID	Short Description	Status
TIR-02-001	The welcome text on the dashboard was not shown when logging in.	Open
TIR-04-001	The system did not display a patient booking success message and produced a database insertion error resulting in patient appointment booking failure when the patient was directly registered.	Open
TIR-10-001	The system allowed Doctor to prescribe duplicate medicine during prescriptions as it didn't prevent duplicate prescription creating within the same appointments.	Open

Document References
HMS_SRS; HMS_TP; HMS_TDS; HMS_TC; HMS_TPS; HMS_TL; HMS_TIR; Katalon Studio execution evidence; Selenium IDE execution evidence; TestCase Studio documentation evidence; Kishan0725 Hospital Management System GitHub repository.

Changes from Plans

All planned testing activities were successfully completed in the prepped local HMS test environment; there has been no significant deviation from the original Test Plan. Testing has been carried out on the local XAMPP server and Google chrome web browser, using both manual and tool supported testing techniques as planned. Katalon Studio, Selenium IDE and TestCase Studio was used as the automation and supporting tools during the testing activities. The final test results will be written in the Test Log and can be summarized in this Test Summary Report.

Comprehensiveness Assessment

A good test coverage of all the major HMS modules and functional features from F001 to F019 were performed, so the test execution was considered as complete. These modules were covered:

1. User Authentication
 - F001 Register Account
 - F002 Login
 - F003 Logout
2. Patient Functions
 - F004 Book Appointment
 - F005 View Appointment History
 - F006 Cancel Appointment
 - F007 View Prescription
 - F008 Submit Contact Query
3. Doctor Functions
 - F009 View Appointment
 - F010 Prescribe Medication
 - F011 Cancel Appointment
 - F012 View Prescription List
4. Admin Functions
 - F013 Add Doctor
 - F014 Delete Doctor
 - F015 View & Search Doctor List
 - F016 View & Search Patient List
 - F017 View & Search Appointment Details
 - F018 View Prescription List
 - F019 View & Search User Queries

The test case is designed based on the SRS, Test Plan and Test Design Specification. Equivalence partitioning, boundary value analysis, decision table testing, state transition testing, use case testing and error guessing are the methods

used for testing. most of the test cases were successful out of the execution process with 3 error to be seen within the critical process.

Testing Schedule

Testing Activity	Planned Date	Actual Date	Status
Test planning and preparation	07/05/2026	07/05/2026	Completed
Test case design and review	07/05/2026	07/05/2026	Completed
Test procedure preparation	01/06/2026	01/06/2026	Completed
Test execution	01/06/2026	01/06/2026	Completed
Test log preparation	01/06/2026	01/06/2026	Completed
Test incident report preparation	12/06/2026	15/06/2026	Completed
Test summary report preparation	14/06/2026	15/06/2026	Completed

Test Coverage Completeness

Test Document	% Completed	Date of Completion
Test Plan	100	07/05/2026
Test Design Specification	100	07/05/2026
Test Cases Specification	100	01/06/2026
Test Procedure Specification	100	01/06/2026
Test Log	100	01/06/2026
Test Incident Report	100	12/06/2026

Test Case Coverage

Test Coverage Item	Count
Total Test Cases	74
Test Cases Passed	71
Test Cases Failed	3
Test Cases Deferred	0

Defect Severity Summary

The defects found during testing are classified based on the incident impact level recorded in the Test Incident Report.

Severity	Incident Impact Mapping	Count
Fatal	Mission Critical	1
Major	Major	2
Minor	Minor	0
Total	All recorded incidents	3

Result Summary

The functional testing session for the Hospital Management System was completed for all planned HMS functions from F001 to F019. A total of 74 test cases were planned and completed. Out of these, 71 test cases successful and 3 case failed.

The results of the passed test cases indicate that majority of the functions present in HMS were performed with expected results. All of these consisted of appointment validation, log out, viewing appointments, cancellation of appointments, prescription view, patient enquiry submission, doctor appointment view, doctor appointment cancel, doctor prescription list view, and admin record management.

However, three incidents have been recorded during testing. The first error was for patients to log in after registering with Gmail two times. For the second incident it was an error in the database while trying to book an appointment when the patient was directly registered. In the third incident, duplicate information on prescriptions was created for the same appointment ID. The fatal defect is the appointment booking error and it impacts on one of the patients' core workflows, causing a fatal database-related error.

Based on the current test results, this HMS has passed a rather large percentage of its test cases, but is not ready for a final release until there will be no open incidents.

Rationale for Decisions

The decision is based on the pass/fail results, incident status, defect severity, and the importance of the affected workflows.

There were 71 of the 74 test cases that passed but three remain open. One incident is considered Mission Critical because the function for booking appointments creates an error in the database, imperiling the patient's ability to complete a fundamental functionality. Two incidents are classified as Major because they affect authentication consistency and prescription data integrity.

It would not be appropriate for the system to be regarded as being in readiness for launch as long as these bugs persist. This is because these are significant processes and the modules that need fixing are related to key processes such as registration/logging in, making appointments, and prescriptions.

Conclusion and Recommendation Based on Test Result

Conclusion:

The desired functional testing cycle has been achieved with the current version of Hospital Management System. The following set of functions (F001 to F019) for

User, Patient, Doctor, and Admin were tested. The majority of the test cases were successful, with the majority of system features able to provide the expected outcome.

However, the system is not entirely operational as three incidents are remain open. The biggest problem is with the appointment booking incident: the booking event crashed because of a database insert issue. The duplicate prescription issue is also significant because it affects the correctness and consistency of medical records. The incident of login should also be fixed since handling duplicate Gmail could hinder patient access and also provide an inconsistent patient record.

Recommendation:

It is advised that the development team addresses all pending incidents before the start of the next test cycle. This should focus on:

1. Fix duplicate Gmail validation in the registration module.
2. Review and correct the login authentication logic affected by duplicate email records.
3. Fix the appointment booking database insertion logic and patient session handling.
4. Add backend validation to prevent duplicate prescriptions for the same Appointment ID.
5. After finish fix the incident,perform regression testing on F001 Register Account, F002 Login, F004 Book Appointment, and F010 Prescribe Medication.
6. Re-run the relevant automated test scripts and update the Test Log, Test Summary Report, and Test Incident Report after retesting.

Once all open incidents have been fixed and retested, then the HMS could undergo the next round of tests for verification and validation.

Approvals		
Name	Job Title	Signature
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